

ALGOFEED

European Food Waste Prevention and Resource-Efficiency Demonstration Infrastructure

Call: SMP-FOOD-2026-FW-STAKEHOLDERS-PJ

Working document for completing the official Part B template

This text was prepared on the basis of the AlgoFeed, pilot, technological, financial, legal and institutional materials made available.

Editorial principle: repeated sections from earlier materials have been merged; the emphasized statements have been retained; the text is built on concrete project activities, metrics and existing pilot experience.

Important checkpoint before finalisation: the 450–600 tonne target appears differently across several documents. In this working version, at least 450 tonnes is used for the full 24-month demonstration, with 600 tonnes as the stretch target, since this is what the minimum and stretch goals in the project support. The phrasing "450–600 tonnes per year" is therefore not used.

Throughput capacity increased from 4.25 to 18.40 tonnes/month/FTE: +340%.

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ALGOFEED: "LET'S NOT ALLOW FOOD THAT CAN STILL BE USED TO BECOME WASTE."

General objective: the demonstration of a measurable, auditable, preventive food-redistribution infrastructure that can be adapted to other European food chains and civil-society networks.

BASELINE EU PROJECT INTERVENTION LEVEL — LIMITATIONS OF BASELINE — ALGOFEED TECHNOLOGICAL INNOVATION (EU ADDED VALUE)

THE ANSWER: WE SOLVE THE OPTIMISATION PROBLEM OF TIME-CRITICAL FOOD REDISTRIBUTION.

Even alongside current European food-rescue systems, a significant amount of consumable surplus is lost because redistribution is not optimised in time, capacity or route.

ALGOFEED: EUROPEAN FOOD-WASTE PREVENTION AND RESOURCE-EFFICIENCY DEMONSTRATION INFRASTRUCTURE.

The narrative of an EU food-waste prevention demonstration infrastructure. AI-based, predictive and logistically optimised redistribution infrastructure.

[ENVIRONMENTAL INTERVENTION POINT]: THIS IS "UPSTREAM FOOD-WASTE PREVENTION."

BASELINE EU PROJECT INTERVENTION LEVEL — LIMITATIONS OF BASELINE — ALGOFEED TECHNOLOGICAL INNOVATION (EU ADDED VALUE)

IDENTIFIED GAP: THE TIME-CRITICAL COORDINATION BETWEEN STILL-CONSUMABLE SURPLUS AND ACTUAL REDISTRIBUTION CAPACITY.

AlgoFeed addresses the coordination gap between surplus food, need, recipient capacity and transport capacity.

AlgoFeed's innovation, in eight points:

1) It pursues a preventive strategy:

Surplus → Consumption, instead of waste — we prevent waste from being created.

AlgoFeed's fundamental difference from traditional food-rescue solutions is that it does not focus on rescuing food surplus after the fact, but intervenes at the very moment the surplus arises.

Integrated into supermarket systems, it can automatically identify and redirect surplus days before expiry. Using artificial intelligence, it organises the delivery of food — often in tonne-scale batches — to appropriate charitable and cold-chain logistics networks.

While consumer food-rescue apps typically try to distribute near-expiry products only a few hours before store closing, AlgoFeed intervenes proactively, integrated into corporate warehouse and inventory-management systems.

2) A financial and sustainability auditing tool:

A digital measurement, monitoring and ESG data-reporting infrastructure.

For large companies, AlgoFeed is not merely a logistics system but also a financial and sustainability auditing tool.

The system calculates and documents the avoided CO₂ equivalent and related environmental impacts. This data can be built into multinational companies' sustainability and ESG reports.

AlgoFeed can be integrated with SAP, Oracle and other enterprise systems, and it operates automatically in the background. Store managers and warehouse staff need no additional administrative work.

3) AlgoFeed's environmental intervention point: the upstream intervention.

AlgoFeed's environmental intervention point is the stage between food surplus and its conversion into waste — we do not want to manage waste better, we want to prevent it from becoming waste in the first place. AlgoFeed therefore does not intervene at waste processing, but one step earlier.

GOAL: Surplus → Consumption, instead of waste — we prevent waste from being created.

This is AlgoFeed's true environmental intervention point: the system intervenes in the process by which food waste is generated, at the point where still-edible food surplus faces the risk of becoming waste.

During the demonstration period, the project will prevent at least 450 tonnes of edible food from becoming waste, measured through digital weighing and e-POD records.

ALGOFEED ADDRESSES WASTE PREVENTION AT THE TOP OF THE WASTE HIERARCHY:

AlgoFeed starts from still-consumable food surplus and keeps it in consumption through intelligent redistribution:

STILL-CONSUMABLE FOOD SURPLUS → INTELLIGENT REDISTRIBUTION → DOES NOT BECOME WASTE

This is a fundamental shift in approach: it does not optimise the utilisation of already-created waste, but prevents its creation.

4) A cross-financed, hybrid business model:

AlgoFeed is not a simple food-rescue app but a preventive logistics platform.

Instead of waiting for food to spoil or become waste, with the help of AI it connects surplus, at the moment it arises, with the nearest and most suitable need.

Companies and retail chains pay for the software, since it provides them with significant waste-management cost savings and ESG-compliance benefits.

From this, AlgoFeed provides a completely free digital infrastructure and intelligent route planning to charitable organisations. The model is therefore simultaneously commercially sustainable and impact-driven.

5) A missing AI and logistics coordination layer:

AlgoFeed provides the missing coordination and logistics layer for already-existing food-surplus donation systems.

It adds an AI-based and logistics-optimisation layer to food-surplus redistribution that can simultaneously take into account: available food stock / need / beneficiary priority / recipient capacity / vehicle capacity / driver availability / cold-chain requirements / and the optimisation of routes and empty kilometres.

6) AlgoFeed is a market-transforming, forward-looking solution that puts food donation on a new footing.

PARADIGM SHIFT: FROM REACTIVE TO PROACTIVE OPERATION.

7) Full automation: transparency and objectivity

AlgoFeed's automated decision support makes redistribution more transparent and fairer. The system prioritises based on objective, measurable criteria, thereby reducing problems arising from bias, lack of information or human decision distortion.

8) Unique and outstanding metrics:

The most important result indicators of AlgoFeed's demonstration project:

- **Logistics response time reduced from 14 hours to 45 minutes**
- **At least 450 tonnes of edible food prevented from becoming waste**
- **340% increase in capacity**
- **Zero-forint NGO infrastructure**
- **AI-based demand forecasting and route optimisation**
- **Reduced empty kilometres and CO₂ load**

These indicators together illustrate AlgoFeed's environmental, social and technological added value.

Attached as annexes: the capacity-growth diagram; the case study of the Vízválasztó Association's test project.

8) Pan-European applicability:

ALGOFEED IS AN INFRASTRUCTURE DESIGNED FOR PAN-EUROPEAN OPERATION;

AlgoFeed was designed for pan-European operation because food waste is a shared challenge across Europe.

A solution tested in one country becomes a pan-European demonstration infrastructure. The platform can adapt to the differing regulatory environments of individual countries while providing a unified digital solution for retail and charitable organisations.

This allows the system to operate effectively in multiple countries and, in the long run, to achieve a significantly broader social and environmental impact.

AlgoFeed's European added value lies in treating the prevention of food waste not as an isolated local solution but as a pan-European, digitally scalable infrastructure.

Building on the pilot already in operation, the model validated during the LIFE demonstration can also be introduced in other Member States and in different retail–charity networks.

ALGOFEED'S CENTRAL QUESTION: WHAT HAPPENS TO EDIBLE FOOD THAT IS STILL LOST DESPITE EXISTING SYSTEMS, BECAUSE OF A LACK OF COORDINATION?

AS AN ANSWER: WE SOLVE THE OPTIMISATION PROBLEM OF TIME-CRITICAL FOOD REDISTRIBUTION.

Even alongside current European food-rescue systems, a significant amount of consumable surplus is lost because redistribution is not optimised in time, capacity or route. ALGOFEED: EUROPEAN FOOD-WASTE PREVENTION AND RESOURCE-EFFICIENCY DEMONSTRATION INFRASTRUCTURE.

The narrative of an EU food-waste prevention demonstration infrastructure.

ALGOFEED: AI-BASED, PREDICTIVE AND LOGISTICALLY OPTIMISED REDISTRIBUTION INFRASTRUCTURE.

ENVIRONMENTAL INTERVENTION POINT: THIS IS "UPSTREAM FOOD-WASTE PREVENTION."

BASELINE EU PROJECT INTERVENTION LEVEL — LIMITATIONS OF BASELINE — ALGOFEED TECHNOLOGICAL INNOVATION (EU ADDED VALUE)

AlgoFeed addresses the coordination gap between surplus food, need, recipient capacity and transport capacity.

IDENTIFIED GAP: THE TIME-CRITICAL COORDINATION BETWEEN STILL-CONSUMABLE SURPLUS AND ACTUAL REDISTRIBUTION CAPACITY.

[How does it differ from other models?]

1) LIFEfoodCycle – AlgoFeed comparison:

LIFEfoodCycle shows that the EU has already funded a digital platform for rescuing food surplus.

ALGOFEED TARGETS A DIFFERENT INTERVENTION LAYER:

WE SOLVE THE OPTIMISATION PROBLEM OF TIME-CRITICAL FOOD REDISTRIBUTION. HOW DO WE OPTIMISE THE SIMULTANEOUS COORDINATION OF HUNDREDS OR THOUSANDS OF DYNAMICALLY CHANGING SURPLUS, DEMAND, CAPACITY AND ROUTE DATA POINTS?

This is already a scale problem. It doesn't just say, "Don't throw this food away." Instead: "Where is it needed?" "Who can take it?" "When does it need to be delivered?" "Which vehicle should go?" "By which route?" "How can it be delivered with the fewest empty kilometres?"

ALGOFEED SUPPORTS THESE DECISIONS WITH INTEGRATED AI-BASED ALLOCATION AND VRP-BASED ROUTE OPTIMISATION.

2) AlgoFeed and LIFEfoodCycle comparison:

Core problem:

LIFEfoodCycle: Rescuing as large a share of food surplus as possible by connecting supermarkets with social organisations.

AlgoFeed: The optimal distribution and delivery of available food surplus to where it is actually needed.

Main technology:

LIFEfoodCycle: A digital platform for managing food surplus and coordinating and tracking the donation process.

AlgoFeed: AI-based decision support and route optimisation, which also takes into account stock, need, priorities, vehicle capacity and available drivers.

3) AlgoFeed and Foodprint comparison:

While Foodprint addresses the awareness and behavioural gap behind food waste, AlgoFeed addresses the coordination gap between surplus food, need, recipient capacity and transport capacity.

One of Foodprint's main problems was how to change actors' behaviour and how to encourage them to prevent food waste.

ALGOFEED: WE MAKE THE PHYSICAL REDISTRIBUTION OF FOOD ALREADY RESCUED BY EXISTING FOOD-DONATION SYSTEMS MORE EFFICIENT.

AlgoFeed: forecasts need → prioritises beneficiaries → optimises allocation → plans deliveries → optimises vehicle capacity → reduces empty kilometres → measures logistics CO₂ impact.

3) AlgoFeed and LIFE Waste2Protein comparison:

LIFE Waste2Protein creates a new product from material that has already become waste or organic residue.

ALGOFEED: WE DO NOT LET STILL-CONSUMABLE FOOD BECOME WASTE. WE DELIVER THE FOOD, WHILE STILL CONSUMABLE, TO THE APPROPRIATE BENEFICIARY.

Main environmental impact:

LIFE Waste2Protein: Reduces waste and provides a sustainable alternative to soy- and fishmeal-based animal feed.

AlgoFeed: Prevents food waste while reducing the energy and CO₂ requirements of the transport needed for redistribution.

The most important difference:

LIFE Waste2Protein: "Once waste has been created, let's turn it into a valuable resource."

ALGOFEED: "LET'S NOT ALLOW FOOD THAT CAN STILL BE USED TO BECOME WASTE."

LIFE Waste2Protein: Circular economy + waste valorisation + resource efficiency

AlgoFeed: Circular economy + food waste prevention + AI digitalisation + resource optimisation + logistics efficiency + CO₂ reduction

Comparison: Inofeed

Inofeed already reuses material treated as biowaste. AlgoFeed intervenes one step earlier than this: it does not allow consumable food to become waste.

It is not just the quantity of waste that must be shown, but the lost resources and their environmental consequences.

THIS IS "UPSTREAM FOOD-WASTE PREVENTION."

SURPLUS → CONSUMPTION, INSTEAD OF WASTE.

ALGOFEED: EUROPEAN FOOD-WASTE PREVENTION AND RESOURCE-EFFICIENCY DEMONSTRATION INFRASTRUCTURE.

BASELINE EU PROJECT INTERVENTION LEVEL — LIMITATIONS OF BASELINE — ALGOFEED TECHNOLOGICAL INNOVATION (EU ADDED VALUE)

4) Munch = AlgoFeed comparison:

Food for Free = digital access for people in need, FoodCloud = digital redistribution, Foodsharing = community rescue, while AlgoFeed = AI-based, predictive and logistically optimised redistribution infrastructure.

THE DECISIVE DIFFERENCE:

Most predecessor systems: "There is food surplus → let's find someone to take it."

These already solve certain sub-problems; AlgoFeed demonstrates the next coordination layer.

AlgoFeed's concept:

"We know in advance where, when and how much surplus will arise → we know in advance where it is needed → we align capacities → AI determines the optimal allocation → VRP determines the optimal delivery."

THIS IS A SYSTEM-LEVEL DIFFERENCE.

Existing systems have already proven the feasibility of food rescue and digital redistribution. AlgoFeed addresses the coordination gap: it forecasts the location and timing of surplus and need, aligns them under capacity and time constraints, and then optimises the physical redistribution.

This is not a national pilot but a measurable, further-developable food-waste prevention model that can also be adopted in other European regions.

With this support, we will develop it into a European demonstration infrastructure that prevents the destruction of still-consumable food surplus, while reducing the resource and CO₂ requirements of redistribution through AI-based demand forecasting and logistics optimisation.

What distinguishes AlgoFeed from traditional food-rescue apps?

AlgoFeed's breakthrough significance shows above all in its pursuit of a preventive strategy.

Within supermarket systems, it automatically redirects surplus days before expiry, thereby:

SAVING FOOD FROM BEING DISCARDED AND REDUCING COMPANIES' UNNECESSARY COSTS.

Traditional, mainly consumer-facing food-rescue apps typically try to get near-expiry food to consumers only a few hours before store closing. AlgoFeed, by contrast, integrated into supermarkets' warehouse systems, uses artificial intelligence to reorganise surplus proactively and in large batches into charitable and cold-chain logistics networks.

In the EU food supply chain, a significant amount of food still fit for human consumption becomes waste due to gaps in the coordination between available surplus, demand and redistribution capacity. The problem is particularly significant in retail and food-distribution processes, where the short shelf life, changing stock levels, limited NGO capacity, and the lack of timely alignment of transport options mean that food surplus can quickly lose its redistributability.

Root cause of the problem:

Current donation and redistribution systems are typically reactive. Available food surplus, local need, the capacity of recipient organisations, and available transport capacity are not aligned in a unified, real-time decision-support system.

Verifying food-safety compliance and tracking shipments can still represent a significant administrative burden.

Even alongside current European food-rescue systems, a significant amount of consumable surplus is lost because redistribution is not optimised in time, capacity or route.

ALGOFEED: EUROPEAN FOOD-WASTE PREVENTION AND RESOURCE-EFFICIENCY DEMONSTRATION INFRASTRUCTURE.

The narrative of an EU food-waste prevention demonstration infrastructure. AI-based, predictive and logistically optimised redistribution infrastructure. Environmental intervention point: this is "upstream food-waste prevention."

BASELINE EU PROJECT INTERVENTION LEVEL — LIMITATIONS OF BASELINE — ALGOFEED TECHNOLOGICAL INNOVATION (EU ADDED VALUE)

AlgoFeed addresses the coordination gap between surplus food, need, recipient capacity and transport capacity.

Root cause of the problem (repeated):

Current donation and redistribution systems are typically reactive. Available food surplus, local need, the capacity of recipient organisations, and available transport capacity are not aligned in a unified, real-time decision-support system.

Verifying food-safety compliance and tracking shipments can still represent a significant administrative burden.

This is AlgoFeed's true environmental intervention point: AlgoFeed intervenes in the process by which food waste is generated, where edible food surplus faces the risk of becoming waste. For this, AlgoFeed provides a digital, AI-based coordination and logistics layer.

Scale and ambition:

Solution level: upstream food-waste prevention + AI-based redistribution + logistics optimisation.

AlgoFeed's goal is not the processing of already-created food waste, but preventing still-consumable food from becoming waste.

Direct environmental goal over the 2-year demonstration period:

- saving approximately 450–600 tonnes per year of food fit for human consumption from becoming waste;
- a 25–30% reduction in surplus/empty kilometres through AI/VRP-based route optimisation;
- increasing the utilisation of transport capacity;
- reducing fuel consumption and CO₂ emissions associated with redistribution.

Measurement and validation:

Food-waste prevention:

1) All rescued food quantities are recorded with digital weighing and e-POD (electronic Proof of Delivery) data.

Based on the data generated by the system, the following can be determined:

- rescued food in tonnes;
- number of redistributions;
- number of beneficiary organisations;
- quantity of waste prevented.

Environmental impact:

AlgoFeed measures two separate impacts:

- 1) the avoided environmental burden from food-waste prevention;
- 2) the avoided environmental burden;
- 3) the direct fuel and CO_{2e} savings from optimised deliveries.

Logistics measurement:

The system compares routes before and after optimisation based on distance travelled, vehicle-capacity utilisation, empty kilometres, fuel consumption and CO_{2e}.

Food safety and traceability:

For chilled/frozen deliveries, IoT/BLE temperature monitoring and digital delivery documentation ensure traceability in line with the specific food-safety requirements.

Economic sustainability:

The system creates economic value for donors:

- reduces costs related to destruction and waste management;
- reduces the administrative burden of redistribution;
- improves the utilisation of logistics capacity;
- provides automated environmental and logistics data to support corporate ESG/CSRD reporting.

Operating model: following the demonstration phase, AlgoFeed will move to a SaaS-based business model.

Minimum demonstration goal:

In the 2025 TRL6 pilot, the automated system achieved an average of 2.5 minutes from signal to matching decision. The project applies a more conservative 45-minute end-to-end operational target, which already takes into account the necessary verification, organisational and physical hand-over processes.

- **2.5-minute pilot response time**
- **45-minute project target**
- **+340% capacity**
- **–82.9% empty kilometres in the pilot**
- **–28% total network kilometres**
- **18.45 tonnes in 3 months**
- **≥450 tonnes in 24 months**
- **600-tonne stretch target**
- **15 stores × 35 kg × 300 days × 2 years = 315 tonnes**

With gradually expanding partner and redistribution capacity during the project, the goal is:

≥450 tonnes of rescued food over the entire demonstration period.

Goal: with full utilisation of the pilot's capacity, expansion of the partner network and optimisation of operations:

600 tonnes of rescued food.

KPI minimum: 450 tonnes.

Stretch target: 600 tonnes.

AlgoFeed – CO₂e environmental impact measurement

Emission source	Type / Scope	Methodology / Source	CO ₂ e savings (3 yrs)	Measurement tool
(Avoided) CO ₂ e/kg	Scope 1 / Scope 2	Defra / EU GREET (0.21 kg CO ₂ e/km)	1.41 t CO ₂ e	GPS / telematics logs
TOTAL	Combined			

Emission / impact source: Food-waste prevention

Impact type: Avoided emissions

Methodology: Food life-cycle based calculation, documented and validated during the project with emission factors.

Initial estimate: 2.5 kg CO₂e/kg of food, to be refined and validated during the project based on the actual composition of food categories.

2-year goal: for 450 tonnes of rescued food, a preliminary estimated ~1,125 t CO₂e of avoided emissions.

Measurement basis: digital weighing data + e-POD + food category + quantity + timestamp.

Emission / impact source: VRP-based logistics optimisation

Impact type: Avoided direct transport emissions

Methodology: comparison of baseline and optimised routes calculated from actual GPS/telematics data, using an emission factor applied to the actual vehicle type and energy/fuel consumption.

2-year goal: 25–30% reduction in empty/surplus kilometres.

CO₂e savings: calculated from the actual kilometre savings measured during the project.

Measurement basis: GPS + telematics + route-planning logs + vehicle and fuel data.

Total environmental impact:

- Emissions avoided through prevented food waste
- Direct CO₂e savings from optimised transport

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Project Summary

AlgoFeed – European food-waste prevention and resource-efficiency demonstration infrastructure

AlgoFeed is not just another food-rescue app. It intervenes at a later point in the problem: between still-consumable food surplus and its actual redistribution opportunity. The system handles surplus, need, recipient capacity and transport capacity within a single decision process.

Even under current systems, items remain that fail to reach the appropriate recipient point in time, capacity or route. AlgoFeed builds an AI-based coordination and logistics layer for this. Through an ERP/WMS connection, it identifies stock nearing expiry and the current capacity of recipient sites, then organises redistribution based on AI-based allocation and dynamic VRP route planning.

Surplus → consumption, instead of waste.

The goal of the 24-month demonstration is to validate the operational prototype in an industrial environment and to advance it from TRL 6 to TRL 8. The project builds on an existing pilot and aims to create a model that can also be adapted in other Member States following the demonstration.

Sole beneficiary: Családokért 2023 Alapítvány (PIC: 864202586).

AlgoFeed is the name of the project and technology, not a separate legal entity.

SPAR Hungary appears as an external supporter and cooperating party linked to the demonstration; it is not a consortium partner and not a co-funder.

1. RELEVANCE

1.1 Background and general objectives

There are already functioning digital and civil-society solutions in European food rescue. However, the problem is not solved simply by recording food surplus somewhere, or by a donor and a recipient organisation making contact. Physical redistribution is time-critical.

Shelf life shortens. Meanwhile, stock, need, recipient-site capacity and vehicle availability all change. If this data remains separate, still-consumable food can lose its redistributability.

AlgoFeed intervenes at this point. It does not optimise the processing of already-created waste. It tries to prevent the still-consumable surplus from becoming waste in the first place.

The system can be connected to corporate stock- and warehouse-management systems. In the project, integration with SPAR Hungary's live data structures is one of the key steps of the demonstration. The technological goal is to validate the existing prototype in an industrial environment, and then develop a fuller, auditable version of the system.

General objective: the demonstration of a measurable, auditable, preventive food-redistribution infrastructure that can be adapted to other European food chains and civil-society networks.

1.2 Needs analysis and specific objectives

The initial pilot showed a tangible operational problem. In the Vízválasztó Association's 12-week test, conducted between 1 October and 24 December 2025, manual coordination produced 380 km/month/vehicle of empty running, 48% capacity utilisation, a 180-minute average response time, and a 14.2% local loss rate.

In the test environment, AlgoFeed prevented 18.45 tonnes of food from becoming waste over three months. Empty or unproductive running fell from 380 km to 65 km/month/vehicle, total network mileage fell by 28%, load utilisation rose from 48% to 86%, and the hub loss rate fell from 14.2% to 1.1%.

The project's objectives:

- preventing at least 450 tonnes of food fit for human consumption from becoming waste over the full demonstration period; 600-tonne stretch target;
- reducing logistics response time from 14 hours to 45 minutes;
- increasing partner operational capacity by at least 340%, based on the measurement logic validated in the pilot;
- at least a 20–25% reduction in empty/surplus transport kilometres;
- better utilisation of vehicle and recipient capacity;
- connecting digital measurement, e-POD, GPS/telematics and environmental data;
- advancing the system from TRL 6 to TRL 8 within 24 months;
- developing a replication model applicable in other EU Member States within a 6–12-month adaptation cycle.

KPI logic: every main goal has a baseline, a measurement method and a target value. The mass of food rescued is measured via digital weighing and e-POD; logistics changes are measured via GPS/telematics and route-planning logs.

1.3 Complementarity, innovation and European added value

AlgoFeed is not built to replace existing food-rescue systems. It places a coordination layer above them. The difference is not simply that "there is a platform," but that the system simultaneously handles stock, need, priorities, recipient capacity, vehicle capacity, driver availability, cold-chain conditions and routes.

The project's distinctive feature is the upstream intervention. The system does not improve the sale or processing of already-created waste. It tries to keep still-consumable food surplus in consumption.

Compared with existing EU and European models, AlgoFeed targets a different problem layer. LIFEfoodCycle-type solutions demonstrate the connection of donors and social organisations and digital coordination. Foodprint

targets the awareness and behavioural side. Waste2Protein and Inofeed already deal with utilising material that has become waste or organic residue. AlgoFeed intervenes one step earlier.

The question here is no longer "who will take this food?" but: where is it needed, who can receive it, when does it need to be delivered, which vehicle should go, and how can this be solved with the fewest possible empty kilometres?

The European added value lies in reproducibility. The modular, API-based system is suitable for managing multiple food chains, NGO networks and vehicle fleets. The project's goal is not to sustain a system exclusive to Hungary, but to create a demonstration model that can be adopted at EU level.

2. QUALITY

2.1 Concept and methodology

The project's methodology is built on a continuous data flow. It starts from stock and expiry data. This connects to recipient sites' capacity and need, and then to data on vehicles, drivers and time windows. AI-based matching ranks available recipient points. The VRP engine then optimises deliveries under the given constraints.

The platform's main modules: the Food Rescue & Aid Module; the Smart Logistics & VRP Routing Module; and the Auditability, e-POD & ESG Reporting Engine. For chilled and frozen deliveries, BLE/IoT temperature monitoring and digital documentation are connected to the process.

The demonstration is built on three technological phases: TRL 6 (M1–M6) — fitting the prototype to live corporate data structures; TRL 7 (M7–M18) — real network operation and fine-tuning; TRL 8 (M19–M24) — finalising the system, independent result verification and the replication model.

2.2 Consortium structure

The project's sole beneficiary is Családokért 2023 Alapítvány. The Foundation carries out the project's coordination, professional management and financial responsibility. External experts and service providers may be engaged for technological development, cybersecurity, legal/HACCP and data-protection compliance, within the subcontracting framework set out in the project budget.

SPAR Hungary is an external supporter and cooperating party linked to the demonstration; it is not a beneficiary, not a consortium member and not a co-funder.

The available institutional background connects to several European research and innovation areas: Waste4Soil, FOOD4CE and SISTERS, as well as professional experience linked to the NÉBIH Wasteless/"Maradék nélkül" programme. The field operations of Családokért 2023 Alapítvány and the Vízválasztó Association are directly linked to food redistribution and to AlgoFeed's validation.

In the Funding & Tenders Portal, Családokért 2023 Alapítvány appears as the sole beneficiary. Subcontractors and external experts are not beneficiary partners; their tasks and costs must be managed under the relevant cost categories.

2.3 Project team, staff and experts

The project team must cover three professional blocks: project and financial management; technological development and integration; and field demonstration, food redistribution and impact measurement.

- project management and financial coordination;
- AI/VRP and systems architecture;
- ERP/WMS API integration;
- e-POD, telematics and IoT/BLE;
- food safety and HACCP requirements;
- ESG/CSRD data and impact measurement;
- field pilot coordination and stakeholder management.

[TO BE COMPLETED: the names, roles, responsibilities, professional experience and planned project working time of key personnel, drawn from the official CVs.]

2.4 Consortium governance and decision-making

Project governance and decision-making are carried out by Családokért 2023 Alapítvány. Technological, pilot, financial and risk decisions are prepared by the responsible persons designated within the project, involving external experts and service providers as needed. SPAR Hungary may act as a reviewing and cooperating party on practical questions of the demonstration, but holds no decision-making or beneficiary status.

Decisions are made in a documented manner. Development, pilot, financial and risk questions must be recorded at project meetings. Changes must be traceable.

2.5 Project management, quality assurance, monitoring and evaluation

One of the project's strengths is that the technological results are not presented merely as system functions. The project expects measurable operational results. Monitoring is based on comparing baseline data with data collected during the project.

- rescued food: digital scale + e-POD;
- response time: system timestamps;
- empty kilometres and total mileage: GPS/telematics;
- capacity utilisation: vehicle and loading data;
- loss rate: stock and exception logs;
- CO₂e: food category, quantity, life-cycle-based factors, and actual logistics data;
- food safety: BLE temperature data and digital shipping documentation.

During the pilot, the data generated by the system is not used only for after-the-fact reporting. The goal is that, while the system is operating, it is already visible where performance falls short of the target value.

2.6 Cost-effectiveness and financial management

According to the available financial framework, the project's total planned cost is EUR 400,000. The requested EU contribution is EUR 192,600, appearing in the document at a 50% funding intensity. SPAR Hungary's EUR 50,000 cash co-financing is also included in the financial framework.

Main cost groups: personnel costs; subcontracting, in particular independent audit, penetration testing and legal compliance; IoT/BLE and other pilot hardware; cloud infrastructure, field testing and communication.

The cost logic is directly linked to the work packages. Development costs serve the TRL 6→8 transition. Pilot costs are needed for real-world operation. The audit and impact measurement ensure the credibility of the results.

[TO BE VERIFIED: that the EUR 400,000 total cost, the EUR 192,600 EU contribution, the co-financing, and the final cost breakdown of the individual WPs match the detailed budget table in the Funding & Tenders Portal.]

2.7 Risk management

Technological risk: fitting to ERP/WMS systems. This must be addressed in the project's early phase, between M1 and M6, before the wider pilot.

Operational risk: changes in recipient and transport capacity. AlgoFeed's methodology is built precisely to address this: capacities and time windows must be taken into account dynamically.

Food-safety risk: a temperature anomaly must not go unnoticed within the system. According to the legal and technological material, such a case can lock the hand-over process, requiring manual dispatcher override.

Data-security and compliance risk: the handling of corporate and logistics data. For this reason, the project includes auditable digital logging, appropriate data-protection design, and planned penetration testing and legal compliance review.

Replication risk: the legal, data-protection, food-safety and logistics environment differs across Member States. The project addresses this not by assuming a uniform legal environment, but by requiring local validation before further rollout.

3. IMPACT

3.1 Impact and ambition

The project's direct impact is preventing still-consumable food from becoming waste. The goal is at least 450 tonnes over the full demonstration period, with a 600-tonne stretch target.

The impact is not built on a single KPI. Alongside rescued food, we measure logistics response time, empty kilometres, vehicle-capacity utilisation, direct transport CO₂e impact, and the avoided environmental burden linked to food-waste prevention.

Results already measured in the pilot show that the system can increase not only the rescued quantity. In the 12-week test, empty running fell by 82.9%, total network distance fell by 28%, and load utilisation rose from 48% to 86%.

The social impact appears on the NGO side: the system's goal is for professional digital coordination and route optimisation to remain essentially free of charge for them. Corporate customers fund the service from the documented savings and ESG value.

This cross-financed model links environmental results, corporate economic interest, and the growth of nonprofit capacity.

3.2 Communication, dissemination and visibility

Communication does not focus on generic technological messages. The project's concrete results must be presented: how much food was not allowed to become waste, how many empty kilometres were avoided, how capacity utilisation changed, and how the data became auditable.

- pilot results and case studies;
- capacity-growth and logistics indicators;
- EU-level professional dissemination;
- stakeholder consultations with commercial and nonprofit actors;
- ensuring the visibility of the project and EU funding in line with the applicable rules.

[TO BE COMPLETED: the final communication channels, events, target groups, responsible partner and timeline.]

3.3 Sustainability and continuation

After the demonstration, AlgoFeed moves toward SaaS-based operation. Corporate customers fund the platform based on the documented logistics, waste-prevention and resource-saving value. Basic access can remain free for nonprofit organisations.

Technological sustainability is supported by the modular, API-based architecture. The system can be fitted to multiple food chains, NGO networks, vehicle fleets and national environments.

The replication goal is to apply the project's methodology in further EU Member States. The available material envisages a 6–12-month adaptation cycle, with prior validation of local legal, data-protection, food-safety and logistics requirements.

4. WORK PLAN — Work plan, work packages, activities, resources and timing

4.1 Work plan

The 24-month work plan leads from technological fitting, through real-world demonstration, to a validated, replicable system.

- M1–M6: baseline, system fitting, ERP/WMS connections, technical and operational preparation — TRL 6;
- M7–M18: live pilot, multi-site operation, AI matching and VRP fine-tuning, e-POD and ESG data flows — TRL 7;
- M19–M24: system finalisation, independent verification, impact evaluation, replication and sustainability model — TRL 8.

The work packages build on one another. The pilot cannot start without clarifying the necessary data and system connections, baseline, and security conditions. The final impact measurement must use the same measurement bases that appeared in the baseline.

4.2 Work packages

WP1 – Technical integration and baseline

Tasks: establishing the baseline; mapping the corporate data and operational processes of participating companies in the demonstration; establishing ERP/WMS connections; setting up the system environment; establishing measurement and data-management foundations.

Main result: a working integration base and documented baseline for the pilot.

Timing: M1–M6.

WP2 – AlgoFeed AI platform and VRP optimisation

Tasks: development of the AI matching engine; dynamic VRP; capacity and time-window management; cold-chain condition handling; IoT/BLE integration; e-POD and ESG data flows.

The technological goal is not simply a faster algorithm on its own. The system must deliver decisions within real operational constraints.

Timing: M1–M18, then finalisation M19–M24.

WP3 – Multi-site pilot demonstration

Tasks: operating a live pilot involving participating retail stores and nonprofit distribution points; use of the platform in real operation; monitoring of chilled deliveries; managing operational errors and deviations.

The pilot's goal is measurable performance in a real environment. The Vízválasztó test project's experience serves here as a baseline and validation reference, not as the project's final result.

Timing: M7–M18.

WP4 – Impact measurement, replication and ESG integration

Tasks: measuring the quantity of rescued food; calculating CO_{2e} impact; comparing logistics savings; ESG/CSRD data reporting; result verification; EU replication model.

The environmental impact from food-waste prevention and the direct transport CO_{2e} savings must be calculated separately, then presented together within the combined environmental impact.

Timing: M7–M24.

WP5 – Project management, communication and stakeholder cooperation

Tasks: project coordination; financial and administrative management; stakeholder consultations; communication and dissemination; project reporting; EU visibility.

Timing: M1–M24.

Deliverables: to be finalised in the official template, focusing on no more than the maximum number of main deliverables permitted by the call guide. The numbers and deadlines of the work packages, budget and deliverables must match.

5. OTHER

5.1 Ethics

The project's main ethical question is not clinical or human-research in nature. The relevant questions concern data handling, automated decision support, food safety, and the handling of operational data related to the organisations involved.

AlgoFeed's decision support is built on objective, measurable criteria: urgency, recipient capacity, vehicle capacity, route and other operational constraints. The goal is to reduce errors arising from human information gaps or possible bias, not to obscure responsibility.

In the event of a food-safety anomaly, a manual intervention option remains within the system. For chilled/frozen products, BLE temperature data and digital documentation are part of traceability.

[TO BE COMPLETED IN THE FINAL TEMPLATE: the specific data-protection roles, data-processing procedures, retention periods and legal bases, based on the actual system and contractual model.]

5.2 Security

The system handles corporate stock, logistics and digital hand-over data. The project therefore treats technological and organisational security as part of the system design.

- controlled access to the integration and pilot environment;
- a digital, traceable event log;
- documentation of e-POD and digital hand-over;
- secure handling of BLE/IoT data flows;
- planned penetration testing;
- legal and compliance review;
- validation of local data-protection and food-safety requirements before replication.

According to the legal working document, at hand-over a dual digital signature, GPS coordinates, a UTC timestamp and a unique item identifier can be recorded; the e-POD forms an auditable trail. These are to be verified and refined in the final technical and legal specification.

6. DECLARATIONS

This section must not be used to create, in one's own wording, a legal declaration that the funding portal or the official Part B template requests in a separate, specific form. The final document should use the declaration elements that appear in the template.

- [TO BE VERIFIED] Declaration regarding the completeness and accuracy of the information provided in the application.
- [TO BE VERIFIED] Exclusion and conflict-of-interest conditions.
- [TO BE VERIFIED] Declarations regarding funding and double funding.
- [TO BE VERIFIED] Acceptance of the EU funding conditions.
- [TO BE VERIFIED] Other declarations required by the official Part B template.

Chapter 6 must not be replaced with freely worded legal text that departs from the official declaration text of the application system.

Annexes

The main body of Part B should not be overloaded with every piece of background material. Relevant evidence should appear in the appropriate annex or in the portal's separate mandatory upload fields.

- Vízválasztó test project — 12-week TRL 6 case study;
- capacity-growth and operational indicators;
- technological core and system architecture;
- legal liability hand-over, e-POD, HACCP and digital hand-over;
- financial framework and detailed budget — reconciled with the official budget table;
- relevant research and institutional background;
- standard CVs of key project team members;
- latest annual activity reports;
- list of previous projects per the official template;
- other documents mandatorily required by the call or the Funding & Tenders Portal.

Pre-finalisation internal checklist

- 450 tonnes = minimum KPI for the full 24 months; 600 tonnes = stretch target.
- 14 hours → 45 minutes: the same baseline and the same definition in every chapter.
- +340% capacity increase: tied to the measurement methodology of the 12-week Vízválasztó pilot.
- 20–25% minimum vs. 25–30% ambition: it must be clearly stated which is the KPI and which is the project goal.
- Pilot result and project target must not be confused with one another.
- Családokért 2023 Alapítvány must appear everywhere as the sole beneficiary; AlgoFeed as the project/technology name, and SPAR as the external supporter.
- Work packages, deliverables, costs and responsible persons/service providers must match.
- Sections 5.1 and 5.2 must not state more than what the actual system and legal documentation support.
- The official declarations of Chapter 6 should be taken from the current template of the Funding & Tenders Portal.
- It is worth preparing the English submission version only after the Hungarian professional version has been finalised.

The two impacts are calculated separately, then presented together within the combined environmental impact.

AlgoFeed's ESG Data Engine ensures the digital collection, linking and auditability of the data.

...moving to a business model. Corporate customers fund the use of the platform from the documented logistics, waste-prevention and resource-saving value, while basic access can remain free for nonprofit organisations.

This is a cross-financed model that delivers both social and environmental impact.

Scalability and transferability:

AlgoFeed's modular, API-based architecture enables the system's integration with different food chains, NGO networks and logistics infrastructures.

The system's design enables:

- application in multiple countries;
- multilingual operation;
- connecting multiple food chains and NGO networks;
- managing different vehicle fleets;
- adaptation to local data and legal requirements.

Replication target:

Based on the methodology developed during the project, the demonstration's results can be adapted in further EU Member States within a 6–12-month rollout cycle, following prior validation of the local legal, data-protection, food-safety and logistics requirements.

The project's goal is not to create a system operating exclusively in Hungary, but to demonstrate a food-waste-prevention and redistribution infrastructure that is reproducible at EU level.

ALGOFEED – SMP FOOD 2026

Long, detailed budget — tabular PDF

Call: SMP-FOOD-2026-FW-STAKEHOLDERS-PJ

Beneficiary: Családokért 2023 Alapítvány | PIC: 864202586 | Duration: 24 months

Total project cost: €400,000.00 Requested EU contribution: €192,600.00 Own/other funding: €207,400.00

Cost category summary

Cost category	EUR
A. Personnel	120,000.00
B. Subcontracting	105,000.00
C.1 Travel & subsistence	18,000.00
C.2 Equipment	55,000.00
C.3 Other goods, works & services	75,831.78
E. Indirect costs (7%)	26,168.22
TOTAL	400,000.00

1. Detailed budget line items

WP	Category	Item	Qty	Unit	Amount (€)	Justification
WP1	A. Personnel	Project manager and coordination	24	2200	52,800.00	Project management, scheduling, reporting, coordination
WP1	A. Personnel	Technical/pilot coordinator	24	1600	38,400.00	Pilots, technical liaison, field operations
WP4	A. Personnel	Monitoring and ESG/impact-measurement expert	18	1000	18,000.00	KPIs, environmental and operational impact measurement
WP5	A. Personnel	Financial and administrative project work	18	600	10,800.00	Financial records, documentation, administration
WP2	B. Subcontracting	AI/VRP development and system integration	1	70000	70,000.00	External development, API/VRP integration
WP2	B. Subcontracting	Cybersecurity / penetration testing	1	12000	12,000.00	Independent security test and report
WP4	B. Subcontracting	Legal, HACCP and data-protection compliance audit	1	13000	13,000.00	Independent compliance review
WP3	C.1 Travel	Pilot sites and field coordination	1	6000	6,000.00	Domestic pilot travel and related costs
WP4	C.1 Travel	EU stakeholder / replication travel	1	8000	8,000.00	International professional consultations and replication
WP5	C.1 Travel	Professional events / dissemination	1	4000	4,000.00	Professional events, presentations
WP1	C.2 Equipment	BLE temperature and cold-chain sensors/gateways	1	20000	20,000.00	Pilot cold-chain and traceability infrastructure
WP2	C.2 Equipment	Mobile/rugged pilot devices	1	10000	10,000.00	Field data entry and e-POD use
WP3	C.2 Equipment	Telemetry and network equipment	1	15000	15,000.00	Pilot data connectivity and measurement
WP4	C.2 Equipment	Measurement / validation hardware	1	10000	10,000.00	KPI and environmental impact measurement
WP1	C.3 Other goods/services	Cloud infrastructure and data-management services	1	24000	24,000.00	24 months of cloud, storage, data management
WP2	C.3 Other goods/services	Software licences, API and monitoring services	1	15000	15,000.00	Services required for technical operation
WP3	C.3 Other goods/services	Field-pilot consumables and operating materials	1	10000	10,000.00	Consumables needed for pilot implementation
WP5	C.3 Other goods/services	Communication, graphic design and translation	1	12000	12,000.00	EU visibility, dissemination and multilingual materials

WP4	C.3 Other goods/services	Data processing, measurement and documentation services	1	8000	8,000.00	Services linked to impact measurement and replication
WP5	C.3 Other goods/services	Project communication and event materials	1	6831.78	6,831.78	Professional presentations and project communication

Total direct costs = SUM(F4:F23)

Indirect cost (7%) = F24 * 7%

TOTAL BUDGET = F24 + F25

2. Cost allocation by work package

WP	Personnel	Subcontracting	Travel	Equipment	Other	Direct total
WP1	30,000	15,000	3,000	12,000	10,000	70,000
WP2	40,000	65,000	2,000	18,000	18,000	143,000
WP3	25,000	18,000	8,000	20,000	22,000	93,000
WP4	15,000	7,000	3,000	3,000	12,000	40,000
WP5	10,000	0	2,000	2,000	13,831.78	27,831.78
TOTAL	120,000	105,000	18,000	55,000	75,831.78	373,831.78

Direct costs: €373,831.78. Indirect cost 7%: €26,168.22. Total cost: €400,000.00.

3. Financing plan

Source	EUR
EU contribution	192,600.00
Eligible personnel cost covered from the Foundation's own resources	120,000.00
Further own / other financial resources	87,400.00
Third-party financial contribution	0.00
In-kind contribution free of charge	0.00
TOTAL FUNDING	400,000.00

Note: the €120,000 personnel cost is part of the eligible project cost; within the financing plan it is the portion covered from own resources. This is not voluntary work and not a separate cost line.

4. Budget control notes

The detailed table was prepared for the 24-month project period and breaks costs down by work package and cost category. The detailed budget does not include D.1 financial support to third parties; the indirect cost was calculated as a 7% flat rate.

This PDF is the long, tabular working version of the detailed budget. For the official Funding & Tenders Portal upload, the application platform's current official Detailed Budget Table format is authoritative.